1. Introduction and business context

Aurora Fashion Group is a mid-scale fashion retail chain with both physical stores and a growing online shopping site. As the retail industry is becoming more and more data intensive, otherwise Aurora Fashion Group's executives understand that using business analytics is a decisive factor in staying ahead mckinsey.com. The fashion retail industry has been faced with some uncertainties of consumer demand, particularly post pandemic, which is why it's important to be equipped with relevant and timely data to respond to that effect mckinsey.com. The company's key focus areas include accelerating revenue growth, unlocking profits, monitoring the performance of stores at the regional level, and finding a balance between online and offline channels. To address such needs, the company, Aurora, develops a business analytics solution to give the company's most senior management team an integrated real time dashboard of key performance indicators. This system specification defines the design of an executive dashboard & analytics solution that matches the needs of the company. The aim of such development is to allow decision making by the CEO, CFO, & Marketing Head of the company to focus on informed strategies based on accurate, factual foundations of business performance to grow business & maximize profitability while remaining supple & nimble in a fast-changing fashion business environment. As goes the saying, "what gets measured gets managed"geckoboard.com.

2. Executive positions & stakeholders

The major executive stakeholders are the CEO, CFO, and Head of Marketing, who need information in these ways:

- **CEO:** Demands an overall view across the overall health of the company, such as (general profitability, Total Revenue, and regional performance). The CEO cares about growing revenue and regional expansion; this dashboard enables easy detection of progress toward Sales objectives and which regions may not be meeting or may be standing out geckoboard.com.
- **CFO:** Focuses on the financials. The important stuff is Gross Profit and Profit Margin. Also reviewing revenues by region and by product line. The CFO is looking at the dashboard to see how the company is making money (margin by region/channel), and that the growth in revenue is feeding bottom line profits phoenixstrategy.group.
- **Head of Marketing:** Focused on sales by channel and by category of goods sold, and how marketing activity affects such sales. The marketing chief looks at Online vs. In Store Orders to measure channel success and Units Sold by Category to measure category success. The data helps direct marketing strategy e.g., shifting budget to online marketing if online sales are taking off, or promoting underperforming categories in a region to drive overall sales growth.

By structuring the system around these roles, every KPI has an "owner" in the executive team (the profit margin belongs to the CFO). Accountability increases in this way. A single source of truth in the dashboard promotes an understanding of data among the executives.

3. KPIs and strategic alignment

Aurora's business analytics platform will target and focus on key performance metrics that are very interrelated to the organization's strategies and goals. Table 1 gives a brief description on the most significant key performance measures, their implications, and their linkages to the organization's strategic plans.

Table 1. Aurora Fashion Group's Key Performance Indicators and Alignment

KPI	Definition/Calculation	Strategic Priority	Executive Owner
Total Revenue	Sum of all sales (£) in the period (all channels).	Revenue Growth	CEO
Gross Profit	Total Revenue – COGS (cost of goods sold).	Profit Margin Improvement	CFO
Profit Margin	Gross Profit / Revenue × 100% (percentage).	Profit Margin Improvement	CFO
Revenue by Region	Total revenue by each region (bar chart).	Regional Performance	CEO
Units Sold by Category	Total units sold by product category (bar chart).	Revenue Growth	Head of Marketing
Revenue Over Time	Revenue trend over time (line chart).	Revenue Growth	CEO/CFO
Online vs In-Store	Online orders vs. store orders (column chart).	Channel Performance	Head of Marketing

Aurora has been able to identify a specific set of KPIs that align with specific strategies. This is a combination of financial and non-financial measures, in a manner similar to a Balanced Scorecard approach balancedscorecard.org. For example, Revenue Growth is tracked through Total Revenue overall, as well as the Revenue Over Time trend, while Profit Margin Improvement is tracked through Gross Profit and Profit Margin percentages, ensuring efficiency is in place for cost control. Regional Performance is then tracked through the Revenue by Region graph, while for Channel Performance, comparisons can be seen through orders by either online or physical channels. Overall, these metrics give a well-rounded view, ranging from financial performance through market and finally channel information, enabling executives to "connect the dots" between any given activities and set goals balancedscorecard.org. Every KPI gets an executive sponsor, as illustrated in the table, making sure there is accountability. When a goal is met or missed, the responsible executive immediately understands its relevance to the business and acts accordingly.

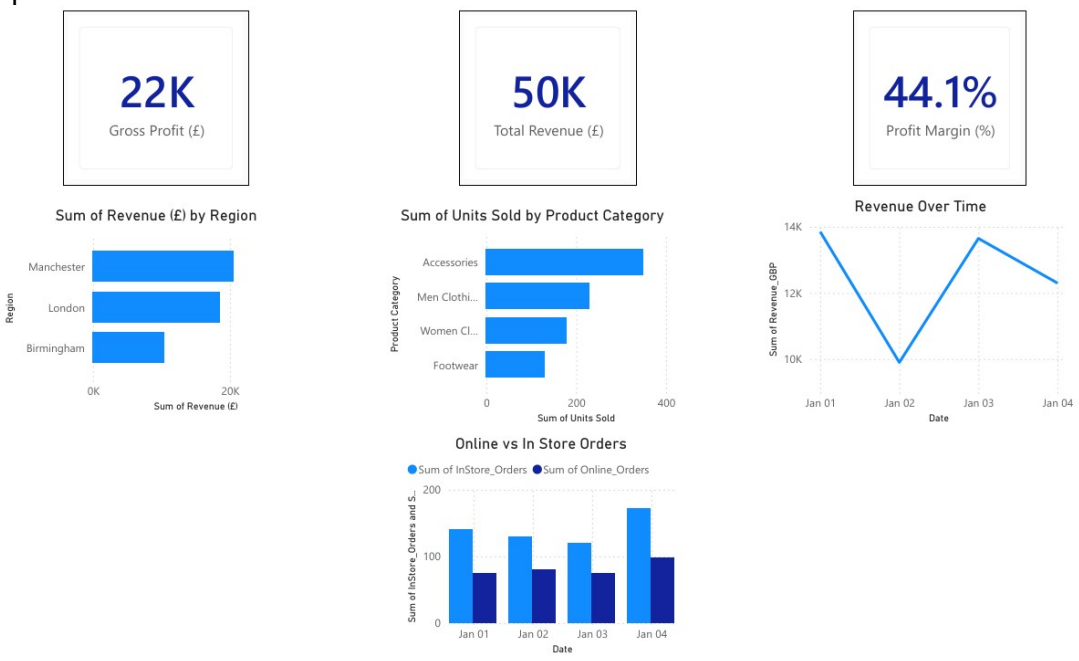
4. Data sources and analytics system architecture

The analytics platform utilizes a variety of data inputs: the point-of-sale systems at Aurora's stores for in store sales data, the e-commerce platform for online sales data, the product and inventory databases for product information and costs, and the marketing/CRM systems for campaign expenditures and customer information. The data inputs for the analytics platform come from an ETL process that consolidates the information into a centralized data warehouse sap.com. The data warehouse also helps combine data related to the current as well as past information from all over the globe, thereby ensuring consistency with respect to calculations of Total Revenue or Profit Margin.

In addition, an analytics/dashboard layer mostly utilizing the BI tool queries the warehouse to generate the visualizations of the executive dashboard. The architectural design enables the dynamic slicing and dicing of data. For instance, executives can slice data by region and channel, and the system aggregates data on the fly. Through organizing data on suitable dimensions, date, region, product, and channel within the system, it facilitates quick data retrieval for the dashboard. For instance, summing up revenue by region is a simple grouped query. This refreshing schedule of the dashboard will be done on a regular basis, say every day, so that executives are always viewing timely data. Proper data governance and security have been implemented. Further, the CFO, the CEO, and the Head of Marketing will have the same unified dashboard, depending on the case, with permissions as per the roles. Altogether, the above structure will make sure that data from all operations that are performed in Aurora is channelled into one system where the information will be cleaned and available for analysis.

5. Executive dashboard specification

The Executive Overview Dashboard can be defined as a one page view with a visual hierarchy thoughtspot.com It points out some key metrics at the top level and several graphs, all of which are kept at approximately 5-7 key elements thoughtspot.com. It provides a good structure that locates the summarization of KPIs at the top of the page followed by detailed graphs afterwards.



Key Dashboard Components:

- **KPI Summary Cards:** At the top of the page, there are three key performance indicators that highlight the total revenue, gross profit, and profit margin of the business. These are key performance measures that all CEOs need to see. The three performance measures can include slight indicators such as colour change or an upward/downward arrow to show how it performed compared to the target or previous performance.
- **Revenue by Region (Bar Chart):** Bar chart comparing the overall revenue earned in the different large regions (such as London, Manchester, Birmingham). This would enable the CEO to see how the different regions perform comparatively. They can be arranged in descending order.
- **Units Sold by Product Category (Bar Chart):** A bar chart showing how many units have been sold for each category: Men's, Women's, Footwear, Accessories. This is very beneficial for the Head of Marketing as he can know which category his products sell most for example, if the Women's Clothing has the highest sales, then Women's Clothing is the most sold.
- **Revenue Over Time:** Line chart of revenue trend over time (e.g., last 12 months - based on revenue). Trends such as the peak periods for the holidays and a revenue growth pattern from quarter to quarter are shown. It helps to enable management to quickly determine how well the company is meeting growth targets on its revenue.
- **Online vs. In Store Orders (Column Chart):** The graph would be a column chart where online orders would be compared to in store orders. This graph would show the split in sales channels. The Head of Marketing would know how both online and in-store channels are performing regarding balancing online shopping and in store shopping. For instance, when online orders beat in store orders in a specific period, this shows that online shopping is successful.
- **Filters:** Interactive filters, such as drop-down selectors, for Date and Region can be found, mostly at the top of the dashboard. The main use of these filters is for the executives to filter the entire dashboard to a date and region of their choice. For instance, selecting "Q4 2025" or "Manchester region" will show only the cards and graphical plots associated with the selected date and region.
- A major aspect of the dashboard is the experience it provides for the user. Everything about the dashboard is designed for clarity. Consistent colour coding is used across all the graphics. Important details are highlighted in the top part of the graphics where the eye is naturally drawn to thoughtspot.com. These ensure that the CEO, CFO, and head of Marketing can then quickly pinpoint the data that they are interested in, and that each one supports one of these strategic questions, such as, 'Which region is strongest?' or 'Is online sales growing?'" The dashboard allows for all these important pieces of data to be considered simultaneously, giving a "big picture view" for analysing how well Aurora is doing, as well as drilling down for specifics when needed.

6. Analytics types supported

Aurora's dashboard supports different types of analytics for decision making:

- **Descriptive Analytics:** It is focused on what has been happening. The dashboard offers descriptive analytics by highlighting historical and current data on performances (for example, total revenue this month and revenue behaviour) ibm.com. These reports provide the executives with the current state of the company (such as "Total revenue for Q3 was £5 million").
- **Diagnostic Analytics:** Allows discovery of why things have taken place. The system enables diagnostic analysis through drill down and comparisons. For example, if the sales dropped in a region, an executive will be able to filter the dashboard to that region and find which product category lagged, or look online vs. in store orders to find out where the issue was analytics8.com. This is such interactive exploration that enables leaders to investigate root causes of trends or problems. For instance, a reduction in the profit margin can be identified if there is an increase in the cost in a particular category.
- **Predictive Analytics:** It can predict what is going to happen. Although today's dashboard is primarily historical, it can include predictive elements such as trend lines or simple forecasts. For instance, a projection can be added to the revenue over time chart to estimate next quarter sales ibm.com. With this, Aurora would be able to expand its influence further by incorporating predictive models such as revenue/demand forecasting to enable its executives to know how to plan based on the predictions.
- **Prescriptive Analytics:** Provides guidance as to what one should do for optimal results. The dashboard can enable prescriptive insights on simple levels such as reminding managers when a particular KPI goes below a certain level (for instance, below 30%, a notification icon pops up for low profit margin) oracle.com. In the future, more advanced prescriptive analytics tools using AI can be applied to prescribe certain actions (such as offering an incentive discount to encourage higher sales of a certain category). Right now, it is up to human analysis for decisions to be made directly from data analysis, but it provides a foundation for prescriptive analytics by pointing out specific areas that management attention should be drawn to.

7. KPI Calculation and Aggregation Logic

- The following details how each KPI is calculated based on Aurora's dataset, as well as how aggregations work:
- **Total Revenue:** This is a calculation involving addition of all values that are present in Revenue_GBP. For example, if the filter applied in the dashboard is in 2025 Q1, then the Total Revenue is a figure that is determined after a summation of all the sales that were made.
- **Gross Profit:** The Gross Profit is derived by subtracting the sum of total Cost of Goods Sold and total revenue. The Cost_GBP per sale is represented in the dataset. The calculation of the system is the sum of both revenue and cost. Lastly, Gross Profit = Sum of Revenue – Sum of Cost.
- **Profit Margin:** The calculation is done after determining the Gross Profit, divided by the Total Revenue, which is provided in percentages, this is accomplished after the

total revenue and cost values are added up to ensure that the calculation is precise and independent of the level of detail in the total cost and total revenue values. An example of the calculation is presented, showing that the profit margin is 40% when the Total Revenue is £1,000,000 and the Gross Profit is £400,000.

- **Units Sold by Category:** Sum of Units_Sold grouped by Product Category, so, we will be able to evaluate the number of sales for each type of product sold, for example, if there are 500 units of Footwear products sold, then these 500 points will be observed under Footwear in the bar chart.
- **Revenue by Region:** Sum of Revenue_GBP by region. The dashboard queries data on the revenue by region to display the total revenue for each region. In all, the total of regions sums to the Total Revenue for consistency to be maintained.
- **Revenue Over Time:** This is the total revenue across time; for example, monthly. It plots the sums in sequence, for example: Jan, Feb, since it is a line chart that shows the trend.
- **Online vs In-Store Orders:** Sum of online order count vs. In Store order count. The data set differentiates between Online Orders and In Store Orders; therefore, this chart will tally each to highlight the difference. (This would be the number of transactions per method of transaction if the data model included transactions as a dimension.)

All these calculations above are defined in the BI tool as measures; hence, all of them automatically take into consideration any filter that might be placed on them (date filter and region filter). The data model is designed in such a way that double counting would not happen; hence a sale would be counted and documented only once along with all details so that any form of calculation above is done accurately. All formulae would be in one place; hence there is an equal definition by the business per KPI. All these formulae were tested in implementation for accuracy using sample data (for example, that regional sales would amount to a total and that profit margin is as if calculated manually).

8. Visualization and specification of user experience

The user experience of the dashboard is very insightful and actionably driven through insights. The user experience through the dashboard includes a clean display of data, which helps in a quick understanding of data among executives. With user experience, visualizations display some conventions, which include conventions on use through colour in graphs, which requires that every region use the same colour. There is also a distinct font used in the display of figures. This aids in lowering cognitive overload on thoughtspot.com. The most critical information, that is KPI cards appears at the top, and graphs display organizational information related to executives in an organized manner, based on priorities. Interactivity enhances the experience for the users; for example, the executives can click through the elements that make up the figure in the bar for the region graph. This means that the click on the "London" option would enable the users to view the data for London for all the elements shown in the graphical illustrations through filtering in the dashboard. The time and region elements make it easy for the users to select the display through the option that triggers the time update in the dashboard illustration shown. This enhances the functionality in the dashboard to make it flexible for use as analytics medium; for example, the executive can begin with the overall view to filter or drill down to analyse an element especially when there is an element of surprise concerning the sales in one region. The layout is responsive to

be easily seen when projected from typical office screens or tablets; the critical elements can be read when the display is smaller. The design choices for visualization and the user experience in the above dashboard align with the best practices concerning BI design to ensure that Aurora's executives obtain insights efficiently and confidently.

9. Implementation considerations and business value

Implementation considerations: The new system must be implemented in a phase-wise manner by Aurora in a way to give attention to the accuracy of the data and acceptance of the system by all concerned. The key technological process would involve the integration of all sources of information into a common warehouse and developing a process of automatic ETL refresh of the same daily. Data integrity verification procedures need to be built into the system, for example, a procedure that validates reconciliation of numbers from source systems and numbers reflected on the dashboard. In addition, there will be close interaction between the IT and analytics team of Aurora and CEO, CFO, and Head of Marketing of the same company for acceptance of layout of the dashboard and final definitions of the key performance indicators while developing the system. Sufficient training should be imparted to the executives of the company and their analysts to utilize the interactive capabilities of the new system appropriately. The new system should be hosted on a secure and private BI system of the company, thereby strictly controlling and safeguarding all confidential information.

Business value: The analytics solution is expected to provide immense business values as it would enable informed and evidence based executive level decisions. The CEOs of the companies would be able to recognize potential threats and opportunities on a timelier basis compared to before. The CEOs could recognize a decrease in sales occurring in a particular region and develop a strategy to counter it within days. The best retailers have reported improvements in sales and profit margins because of analytics. Research indicates a sales increase of 10% and enhanced profit margins of 5-10% by the use of analytics based informed business decisions on matters related to inventory, pricing, and promotions by the best retailers mckinsey.commckinsey.com. Aurora would also look to see related benefits to revenue growth from the earlier recognition of trends and benefits to margins through the early identification and remedying of inefficiencies. The challenge of where to focus their time and effort would be resolved because all the executives would be focused on the same trustworthy data. Finally, the efficiency derived from the automated visual dashboard ensures less manpower is devoted to report generation, thereby providing ample time for analysis and implementation. In conclusion, the analytics system ensures the Aurora Fashion Group makes informed decisions quicker, right in line with their strategy, thus staying ahead in their fast-moving industry.

10. Conclusion

In conclusion, the new executive analytics tool implemented at Aurora Fashion Group is a well-crafted tool developed to address the company's objectives through data. As discussed in the report, the set of carefully identified KPIs (including revenue growth, profitability, and geographic as well as marketing performance) will be displayed in an organized manner specifically for the CEO, CFO, and the Marketing Head. With the integration approach involving data from various aspects of the company and the best practices followed in the design approach, the tool will help the senior management at Aurora Fashion Group make informed decisions.

Such a business analytics dashboard should become a core tool in strategic management for Aurora, allowing it to interpret raw data into operational intelligence. It is expected to aid management in keeping abreast of progress on set goals in real time, detect potential issues in time, and seize valuable opportunities, whether such a move entail changing course in a nonperforming region or fully exploring a rapidly growing online business segment. In the long run, the adoption of this business analytics system is expected to help Aurora realize its plans for growth.

11. References

- Analytics8 (2025). *What Are the Four Types of Analytics and How Do You Use Them?* Analytics8 Blog updated 1 Sept 2025. Available at: analytics8.com [Accessed 2026].
- Balanced Scorecard Institute (n.d.). *Balanced Scorecard Basics*. Available at: balancedscorecard.org [Accessed 2026].
- Devillard, S., Harreis, H., Landry, N. and Sanchez Altable, C. (2021). *Jumpstarting value creation with data and analytics in fashion and luxury*. McKinsey & Company, 14 Oct 2021.
- Geckoboard (n.d.). *CEO Dashboard Example – Executive Dashboards*. Geckoboard Guides. Available at: geckoboard.com [Accessed 2026].
- IBM (n.d.). *What is big data analytics? – Four data analysis methods*. IBM Think Blog. Available at: ibm.com [Accessed 2026].
- Oracle (2023). *How Data Analytics Informs the Fashion Industry*. Oracle Retail. Available at: oracle.com [Accessed 2026].
- Phoenix Strategy Group (2025). *Top Metrics for CFO Dashboards in 2024*. PhoenixStrategy.group blog, 4 Jan 2025.
- ThoughtSpot (2025). *Dashboard Design: Examples and Best Practices*. ThoughtSpot Data Trends. Available at: thoughtspot.com [Accessed 2026].
- ISYS40251 Deriving Value Data Analytics (2026). Aurora Fashion Retail Sales Dataset. Dataset. Dummy dataset created for academic purposes. Nottingham Trent University. Available at: https://docs.google.com/spreadsheets/d/1AE8aQ_Cw2OwCySS56cX1ST-q31w2NNip. [Accessed 2026].